

KSP Theory Assignment

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1 Introduction

In 1977 the Voyager Probes were launched. Their main mission was to collect data from Jupiter and Saturn; however, they are special for another reason. They are the only spacecraft to operate outside the heliosphere. This remarkable feat was possible due to a special alignment of the planets that allowed the probes to utilise multiple gravity assist maneuvers to propel them at a rapid pace.

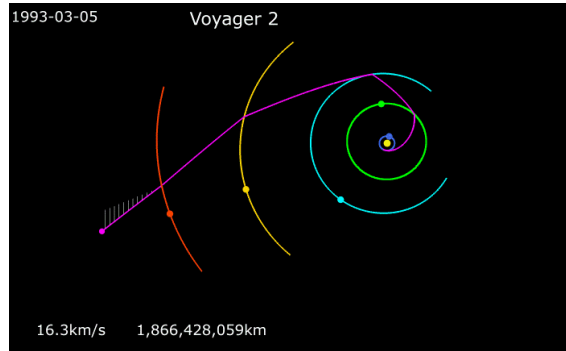


Figure 1: Trajectory of Voyager-2 using 3 slingshot maneuvers

Gravity assist/slingshot maneuvers use the gravity of a massive celestial body to significantly increase the speed of the spacecraft without the use of fuel. When calculating the change in speed (or Δv) we may treat it like an elastic collision since there is no loss of kinetic energy.

1.1 Question 1

Suppose there is a planet of mass M and a spacecraft of mass m , they are moving towards each other and the planet has a velocity of v_p whereas the initial velocity of the spacecraft is v_i . After the slingshot the spacecraft has velocity v_f directly opposite to its initial velocity. What is the relation between v_i and v_f if both velocities are when the spacecraft is far away from the planet. The given velocities are in the solar reference frame. You may make any justified assumption.

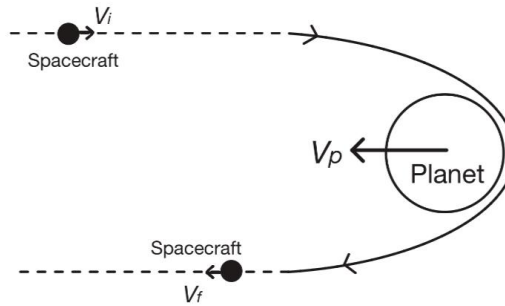


Figure 2: Refer to this figure for Question 1

1.2 Question 2

The situation described in the previous question is not realistic and in reality the spacecraft may approach the planet from any angle relative to it's velocity. Imagine a situation where the spacecraft approaches with a velocity $\vec{v}_i$ that is at an angle θ_i to the velocity $\vec{v}_p$ of the planet and exits with a velocity $\vec{v}_f$ with an angle θ_f to the planet's velocity. Find $|\vec{v}_f|$ in terms of $|\vec{v}_i|$, $|\vec{v}_p|$, θ_i and θ_f considering the given velocities are in the solar reference frame. Also show that we can slow down the spacecraft for "Gravitational braking".

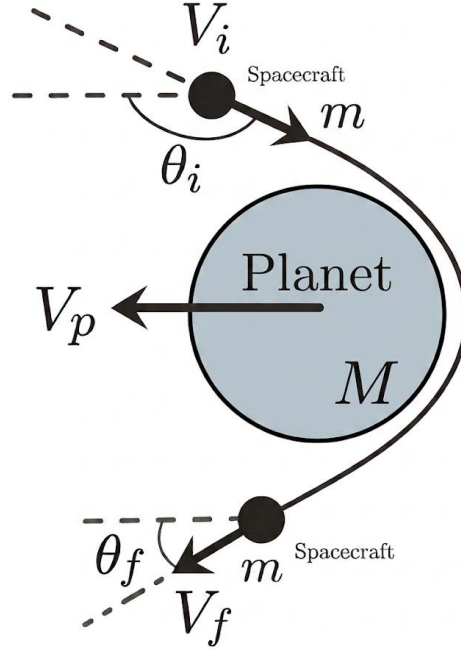


Figure 3: Refer to this figure for Question 2

1.3 Question 3

In the Rutherford Scattering experiment one can calculate the angle of deviation with the initial velocity of the deflected particle, the charge on the deflecting particle and the impact parameter. Similarly one can calculate the angle of deflection where instead of electrostatic force we have gravitational force. Consider a planet of mass M and a spacecraft approaching it with velocity v in the frame of the planet with an impact parameter b where $b >$ radius of the planet. Find the angle of deviation and the distance of closest approach in the planet frame assuming that the spacecraft does not enter the planet's atmosphere.

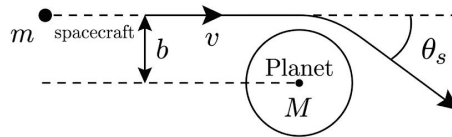


Figure 4: Refer to this figure for Question 3

1.4 Question 4

A Hohmann transfer is an elliptical orbit used to transfer a spacecraft between two circular orbits around a central body (like the Sun). It is one of the most fuel-efficient ways to move between two coplanar circular orbits when the transfer takes the spacecraft from a lower orbit to a higher orbit (or vice versa). In the ideal case with circular orbits the transfer ellipse is tangent to both the inner and outer circular orbits: its perihelion (closest point) touches the inner orbit, and its aphelion (farthest point) touches the outer orbit and it traverses exactly 180 degrees around the primary. Two fuel burns are required, one at the perigee to enter the transfer ellipse and one at the apogee to exit the transfer ellipse at the required radius.

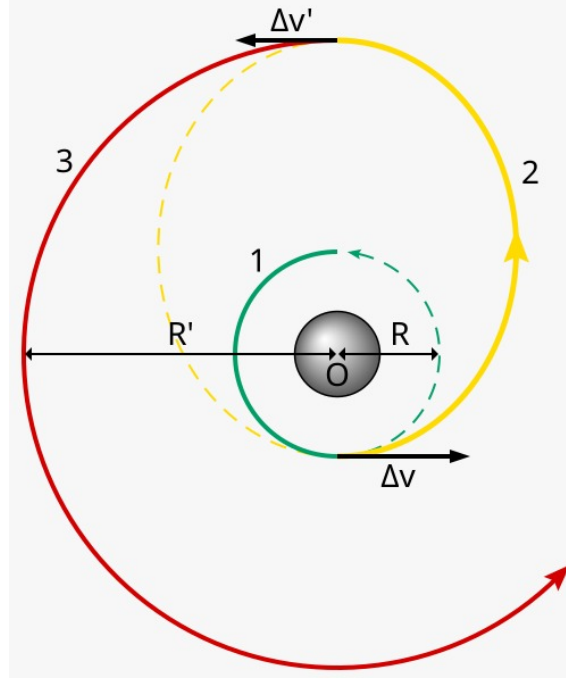


Figure 5: Orbit 1 is Earth's Orbit, Orbit 2 is the transfer ellipse, Orbit 3 is Jupiter's orbit

A spacecraft performs a Hohmann transfer from Earth's circular orbit to Jupiter's circular orbit around the Sun. Both orbits are coplanar. Ignore planetary gravitational effects during the transfer.

(Radius of Earth's circular orbit = R , Radius of Jupiter's Circular Orbit = R' , Gravitational Constant = G , Mass of Sun = M_S)

- Find the velocity changes required at the perigee(Δv) and apogee($\Delta v'$)
- Using Kepler's Third Law find the time taken for the maneuver
- If the spacecraft travels from Earth to Jupiter then find the initial angular separation of the two planets from the sun's frame.

(Note that a proper derivation with suitable diagrams is expected and not just the final answer)

1.5 Question 5

In the previous question we used the Hohmann Transfer to move a spacecraft from Earth's orbit to Jupiter's orbit. Now the spacecraft approaches Jupiter at an angle of 60 degrees with the planet's velocity. The initial velocity v_i of the spacecraft is $36,000 \text{ km/h}$ and the impact parameter is $650,000 \text{ km}$. All of the above observations are in the solar reference frame. Assume circular orbits.

(Gravitational constant(G) = $6.67 \times 10^{-11} \text{ Nm}^2\text{kg}^{-2}$, Mass of Jupiter = $1.9 \times 10^{27} \text{ kg}$, Mass of spacecraft = $50,000 \text{ kg}$, Time period of Jupiter's orbit = 12 years , Radius of Jupiter's Orbit = $7.78 \times 10^8 \text{ km}$, Time period of Neptune = 165 years , Radius of Neptune's Orbit = $4.54 \times 10^9 \text{ km}$)

- (a) Find the distance of closest approach of the spacecraft to Jupiter
- (b) After the spacecraft performs a gravitational slingshot around Jupiter, it reaches Neptune. Assume that the path of the ship after the slingshot maneuver is a straight line. Find the angular separation between Jupiter and Neptune when the spaceship was originally launched from Earth.